# DISCOVERING THE PYTHAGORAS THEOREM!

# MEET OUR SHAPES FRIENDS!

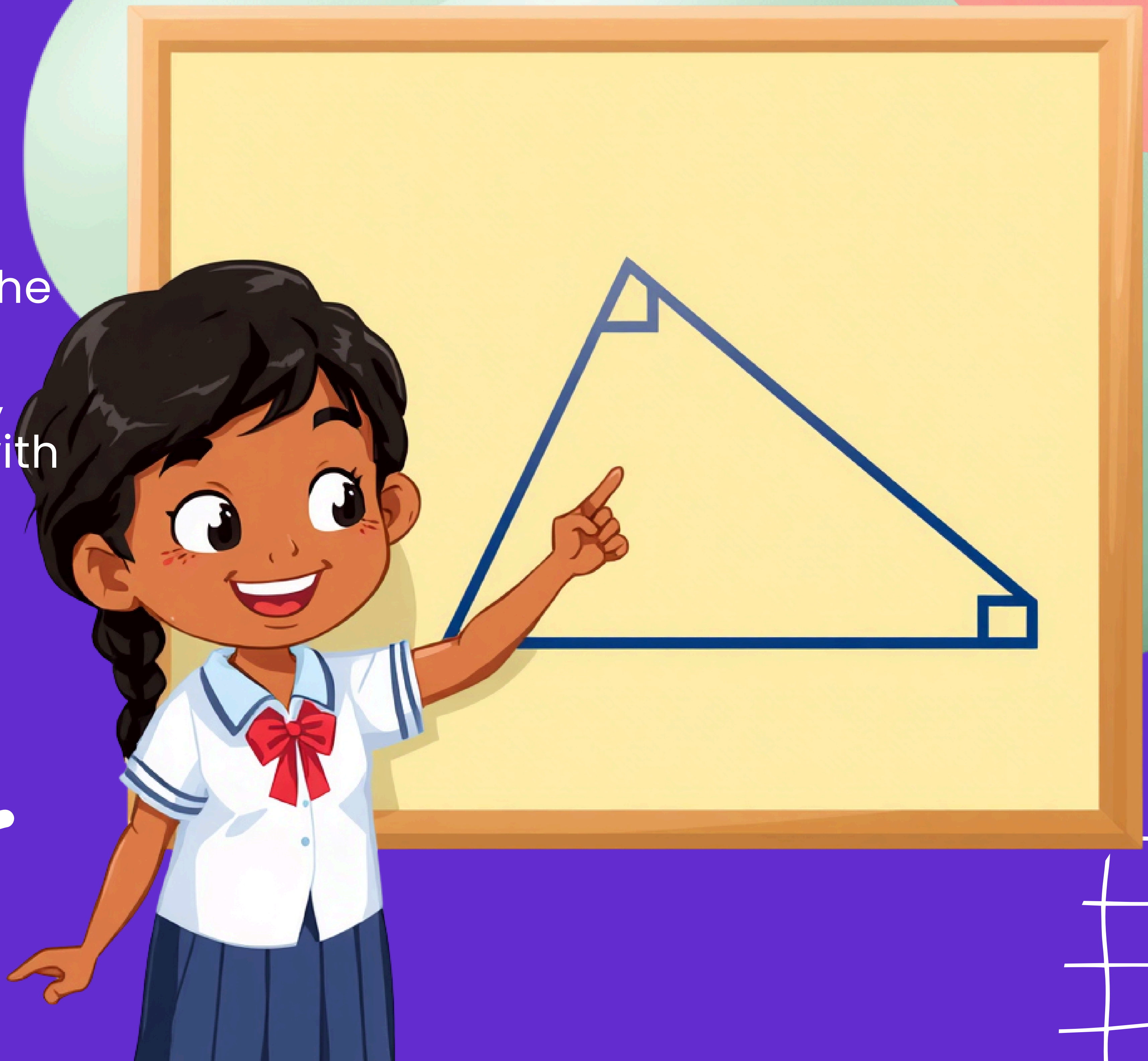
Let's explore some shapes and their special secrets! Triangles, squares, and rectangles are all around us — and they hold amazing maths magic!

Spot the triangle, square, and rectangle around you!

Every shape has sides, corners, and a secret to share!

# WHAT IS A RIGHT-ANGLED TRIANGLE?

A right-angled triangle has one special corner — just like the corner of a book or the edge of your notebook!
That special corner is called a right angle, and it measures exactly 90°. We mark it with a tiny square symbol in the corner.



**Look for the little square — that's your right angle!** 📐

# LET'S LOOK AT THE SIDES!

Every right-angled triangle has three special sides. Each side has its own name and role. Let's learn what makes each side unique!

Hypotenuse: The longest side, always opposite the right angle.

Side A & Side B: The two shorter sides that form the right angle.

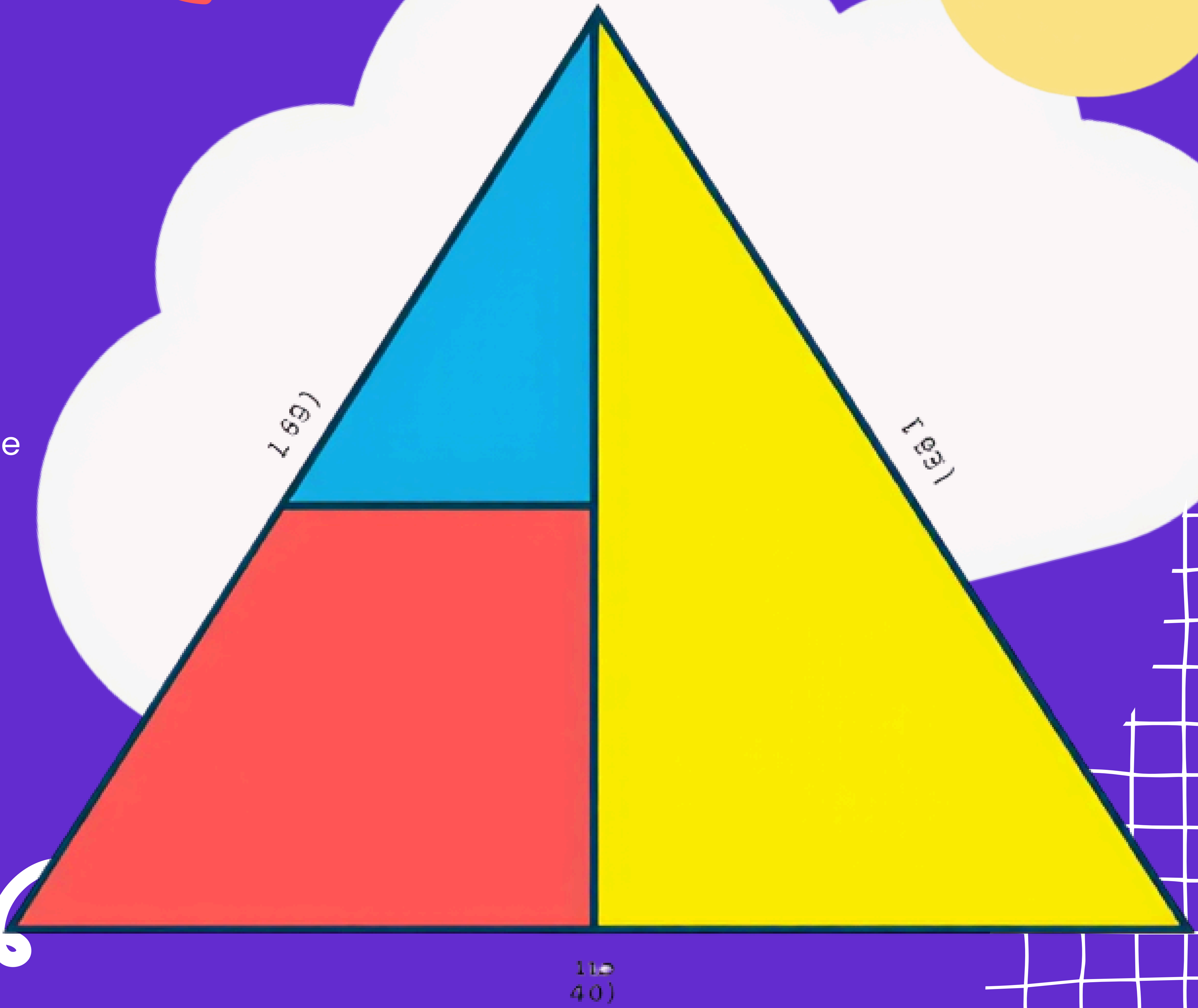
Tip: The hypotenuse is always the biggest side in a right triangle!

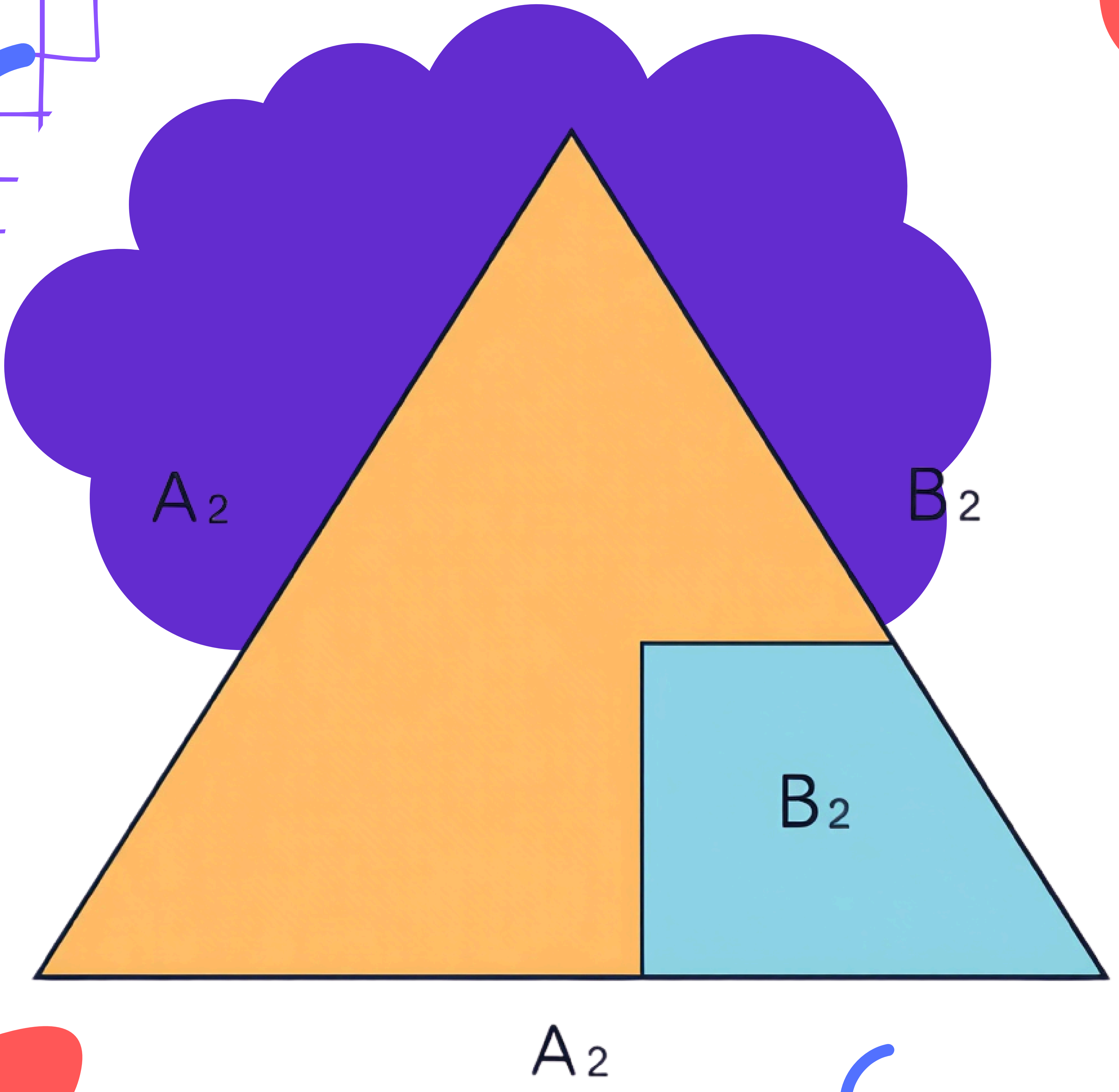
# SQUARES ON EACH SIDE!

## Building Squares

Look! We made squares on each side of the triangle. Each side has its own colourful square — can you see how they fit together?

**The squares on Side A and Side B together equal the square on the Hypotenuse!**






# WHAT DOES THE THEOREM SAY?

The area of the big square equals the sum of the areas of the two smaller squares. In short:
$a^2 + b^2 = c^2$

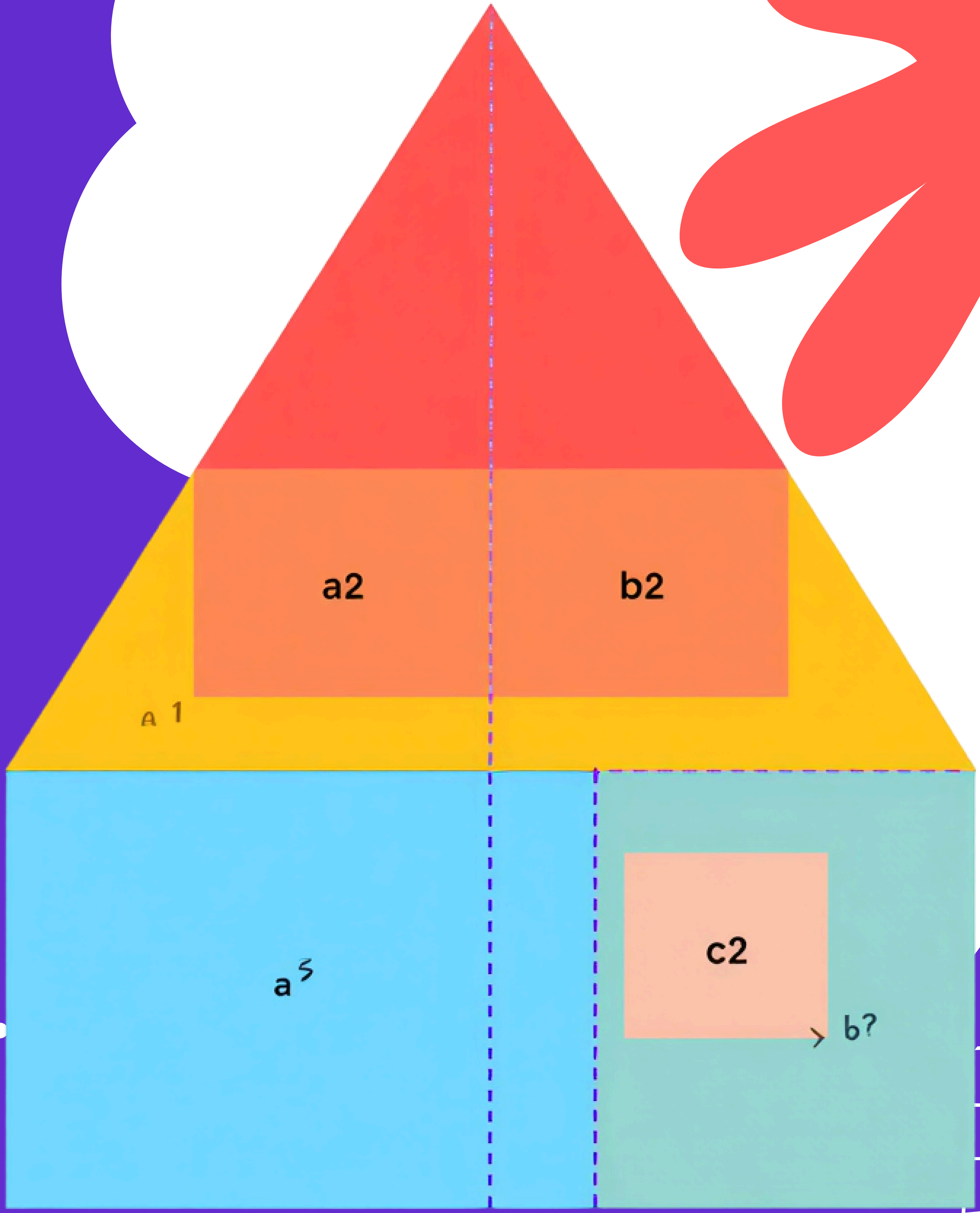
$c^2$ is the biggest square — on the longest side (hypotenuse)!

$a^2 + b^2 = c^2 \rightarrow$ Add the two small squares to get the big one!

# VISUAL PROOF — LET'S SEE!

See how the two small squares fit into the big one?
The square on the longest side ($c^2$) has exactly the same area as the two smaller squares ($a^2$ and $b^2$) added together!
$a^2 + b^2 = c^2$

**The big square = the two small squares combined. Every time!**



# REAL-LIFE EXAMPLE: THE LADDER PROBLEM!

A ladder leans against a wall and forms a right-angled triangle! How far up the wall can the ladder reach? Let's use the Pythagoras theorem to find out.

The ladder is the hypotenuse — the longest side of the triangle!

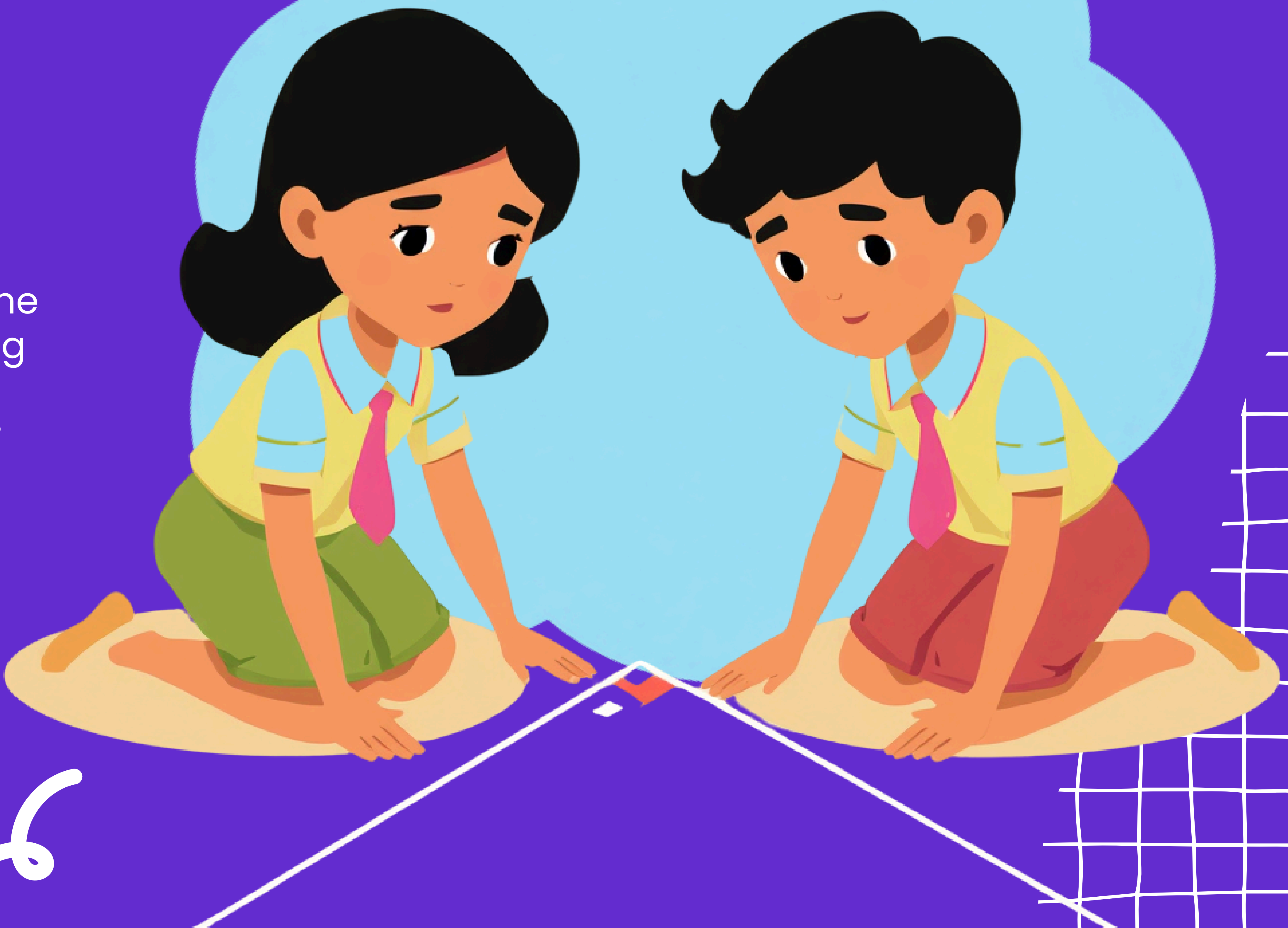
Use $a^2 + b^2 = c^2$ to find the missing side length.

# CLASSROOM EXAMPLE

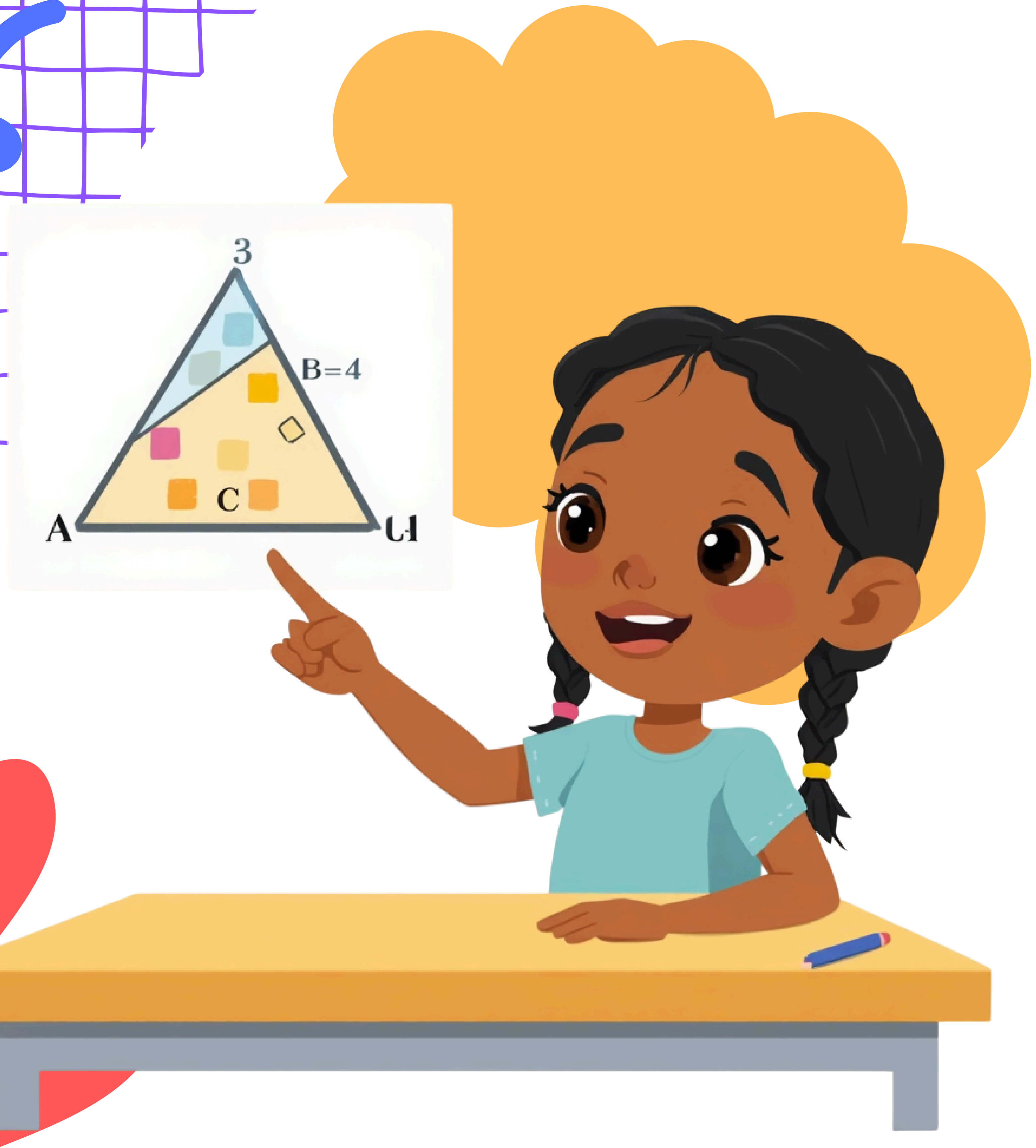
**Measuring on the Floor!**

Riya and Arjun mark a right triangle on the classroom floor. One side is 3 metres long and the other is 4 metres. Can you help them find the length of the longest side?

**What do we need to find?**
**The hypotenuse — the longest side of the triangle!**



# LET'S CALCULATE TOGETHER!




Let's solve a triangle step by step! If $a = 3$ units and $b = 4$ units, we can find the longest side c using $a^2 + b^2 = c^2$

Step 1:
$a^2 = 3^2 = 9$
$b^2 = 4^2 = 16$

Step 2:
$a^2 + b^2 = c^2$
$9 + 16 = 25$

Step 3:
$c^2 = 25$
So $c = \sqrt{25}$
$c = 5$ units! ✓

Answer: The hypotenuse is 5 units long!

# PRACTICE PROBLEMS!

Try these on your own — use $a^2 + b^2 = c^2$ to find the answers. You've got this!

- Find the hypotenuse when $a = 6$ and $b = 8$.
- Find the missing side when $c = 10$ and $a = 6$.
- True or False: A triangle with sides 5, 12, 13 is right-angled?

# FUN QUIZ TIME!

Which side is the hypotenuse? Look at the triangles below and think carefully before you answer!

Triangle A: Sides 3, 4, 5 — Which is the longest side?

Triangle B: Sides 6, 8, 10 — Can you find it?

Triangle C: Sides 5, 12, 13 — Point to the hypotenuse!

Hint: The hypotenuse is always the side opposite the right angle!

# THANK YOU FOR LEARNING WITH US!

**You Did Amazing Today!**

You explored right-angled triangles, met the hypotenuse, and discovered the magic of $a^2 + b^2 = c^2$! Keep practising and you will be a Pythagoras pro!

**Keep curious, keep exploring — maths is full of wonderful surprises!**